

Lecture 2 –System setup for Molecular Dynamics Simulation

We will generate all three configurations using Packmol in a similar arrangement, along with the GROMACS force field.

System label	Electrolyte formulation (mol ratio)	Molecules (L:P)
L1P2	LiFSI + Pyr ₁₃ FSI (1:2)	25:50
L2P2	LiFSI + Pyr ₁₃ FSI (2:2)	50:50
L3P2	LiFSI + Pyr ₁₃ FSI (3:2)	75:50

GROMACS Force Field

```
; created by fftool

[ defaults ]
; nbfunc  comb-rule  gen-pairs  fudgeLJ  fudgeQQ
1          3          yes        0.5      0.5

[ atomtypes ]
; name  at.nr  mass  charge  ptype  sigma  epsilon
Li       3    6.9410  1.0000  A      2.12600e-01  7.64800e-02
FSI      9   18.9980 -0.1300  A      3.11800e-01  2.55400e-01
SBT     16   32.0660  1.0200  A      3.55000e-01  1.04600e+00
NBT      7   14.0000 -0.6600  A      3.25000e-01  7.11280e-01
OBT      8   15.9990 -0.5300  A      3.15000e-01  8.37360e-01
N4       7   14.0070  0.1200  A      3.25000e-01  7.11280e-01
C1       6   12.0110 -0.1700  A      3.50000e-01  2.76140e-01
C2       6   12.0110  0.0100  A      3.50000e-01  2.76140e-01
H1       1    1.0080  0.1300  A      2.50000e-01  1.25520e-01
HC       1    1.0080  0.0600  A      2.50000e-01  1.25520e-01
CT       6   12.0110 -0.1800  A      3.50000e-01  2.76140e-01

[ moleculetype ]
; name      nrexcl
Li+          3

[ atoms ]
; nr  type  resnr  residu  atom  cgnr  charge
1    Li     1    Li+    Li     1    1.000000

[ bonds ]
; ai  aj  func  b0      kb

[ constraints ]
; ai  aj  func  b0

[ pairs ]
; ai  aj  func

[ angles ]
; ai  aj  ak  func  th0      cth

[ dihedrals ]
; ai  aj  ak  al  func  coefficients
```

Simulations Input Parameters

To run a GROMACS MD simulation, we need a job script file and input parameter files, which are given below

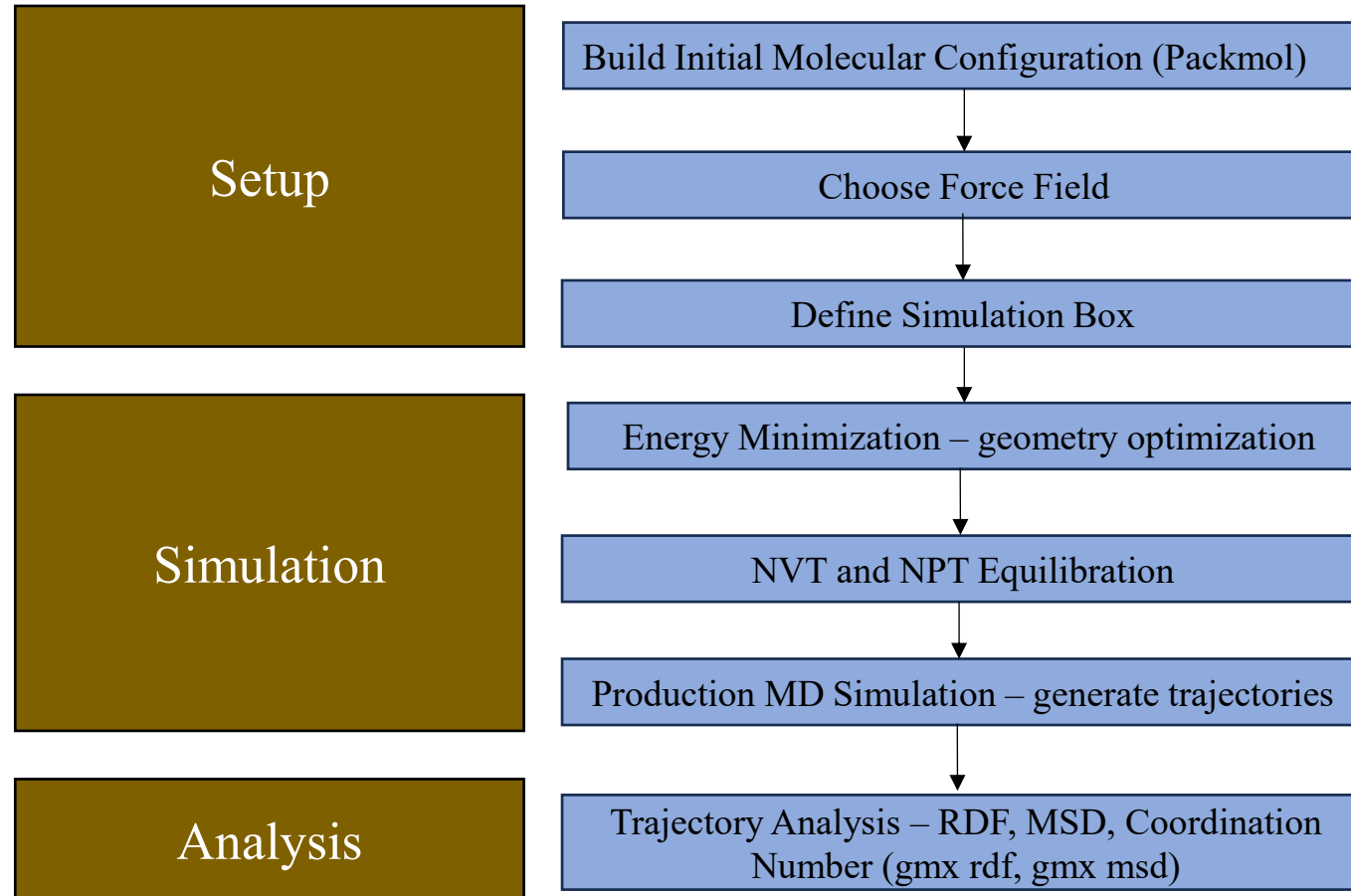
Em → Energy minimization

NVT → Equilibration, Berendsen

NPT → Parrinello–Rahman equilibration

Final → Prod (production MD)

Molecular Dynamics Simulation Framework



Thank you